Learning, Adaptation, and Generalization for Autonomous Agents and Robots in Open-World Environments

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Abstract

Autonomous agents and robots deployed in real-world environments must operate under open-world conditions in which unforeseen changes, novel objects, and unexpected task variations are inevitable. However, most contemporary artificial intelligence and robotics systems are developed under closed-world assumptions where environments, tasks, and dynamics encountered during deployment closely match those observed during training. These assumptions limit the ability of intelligent systems to adapt when prior knowledge becomes insufficient and to generalize robustly across new situations. This dissertation develops methods for enabling autonomous agents and robots to operate effectively in open-world environments through learning, adaptation, and generalization. To this end, it introduces benchmark environments that systematically inject novelty and enable rigorous evaluation of novelty-aware agents. It further develops adaptive planning-learning systems that integrate task and motion planning with reinforcement learning-based recovery mechanisms, allowing agents to detect execution failures and adapt their behavior when previously learned knowledge becomes insufficient. Finally, it demonstrates how robots can acquire generalizable interaction capabilities through object-centric and force-based reinforcement learning policies that enable sustained-contact manipulation across diverse objects and tasks. Together, these contributions provide a unified perspective on building autonomous agents and robots capable of operating in open-world environments. By combining benchmark-driven evaluation, adaptive planning-learning systems, and generalizable embodied interaction policies, this work advances the development of intelligent systems that can robustly operate in complex, dynamic, and evolving real-world settings.

Acknowledgments

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Chapter 1

Introduction

Autonomous agents and robots are increasingly expected to operate in complex real-world environments such as homes, warehouses, hospitals, and urban spaces. In these settings, agents must interact with diverse objects, perform varied tasks, and respond to changing environmental conditions. Unlike controlled laboratory settings, real-world environments are inherently open, new objects may appear, tasks may change, and interactions may produce outcomes that were not anticipated during system design or training. Developing systems that can operate robustly in such environments remains one of the central challenges in artificial intelligence and robotics.

Most contemporary AI and robotic systems are developed under closed-world assumptions, where the environment, tasks, and dynamics encountered during deployment closely match those observed during training. These assumptions simplify learning and planning but limit the ability of agents to operate reliably when confronted with novelty or unexpected situations. In open-world environments, previously learned models may become incomplete or invalid, action execution may fail due to unforeseen circumstances, and interaction policies may not generalize to new objects or tasks. Addressing these limitations requires systems that can adapt when prior knowledge becomes insufficient and acquire behaviors that generalize across diverse interactions with the environment.

This dissertation investigates how autonomous agents and robots can operate effectively in open-world environments through learning, adaptation, and generalization. The work focuses on three interconnected problems. First, developing environments and benchmarks that systematically introduce novelty and enable rigorous evaluation of open-world agents. Second,

enabling agents to recover from failures during task execution by integrating planning with learning-based adaptation. Third, enabling robots to learn interaction policies that generalize across objects, tasks, and embodiment.

The first part of this dissertation addresses the need for systematic evaluation of open-world agents. We introduce benchmark environments that enable controlled injection of novelty and allow researchers to evaluate how agents respond to unexpected changes in their environment. These environments provide a platform for studying adaptive behavior and for comparing different approaches to open-world autonomy.

The second part of the dissertation focuses on adaptive planning-learning systems that allow agents to recover from execution failures. In open-world environments, plans may fail because of incomplete models or unexpected environmental changes. This work develops mechanisms that integrate task and motion planning with reinforcement learning-based recovery strategies, allowing agents to detect failures and adapt their behavior when previously learned knowledge becomes insufficient.

The third part of the dissertation investigates how robots can acquire generalizable interaction capabilities. Real-world robotic tasks often require sustained contact with objects and interactions that differ from those encountered during training. This work explores object-centric and force-based reinforcement learning approaches that enable robots to learn manipulation policies capable of generalizing across objects, tasks and particular robotic embodiments.

Taken together, the chapters of this dissertation develop a unified perspective on building autonomous agents and robots capable of operating in open-world environments. By combining benchmark-driven evaluation, adaptive planning-learning systems, and generalizable embodied interaction policies, this work advances the development of intelligent systems that can robustly operate in the real world.

1.1 Thesis Contributions

This dissertation develops methods for enabling autonomous agents and robots to operate effectively in open-world environments through learning, adaptation, and generalization. The contributions address how such systems can be evaluated under novelty, how they can adapt

when execution failures occur, and how robots can acquire interaction policies that generalize across objects, tasks, and robotic embodiment.

The main contributions of this dissertation are summarized as follows:

- It introduces benchmark environments and evaluation methodologies that systematically introduce novelty into agent environments, enabling rigorous study of autonomous systems operating under open-world conditions. These environments provide mechanisms for controlled novelty introduction and support evaluation of how agents detect, respond to, and adapt to unexpected changes in their environment.

- It develops adaptive planning–learning systems and architectural frameworks that enable agents to recover from execution failures in changing environments. By integrating planning mechanisms with learning-based adaptation, these systems allow agents to detect when previously learned models become insufficient and modify their behavior to continue task execution despite unexpected changes.

- It introduces object-centric reinforcement learning approaches for robotic interaction that learn manipulation policies in force space. These methods enable robots to acquire interaction strategies that generalize across objects, tasks, and robot embodiments, supporting manipulation behaviors involving sustained contact and diverse physical interaction dynamics.

1.2 Thesis Outline

The remainder of this dissertation is organized as follows.

Chapter 2 presents background and related work relevant to open-world autonomy. It discusses the limitations of closed-world assumptions in contemporary artificial intelligence and robotics systems and reviews prior work in task and motion planning, reinforcement learning, novelty-aware agents, and robotic manipulation. The chapter establishes the conceptual foundations used throughout the dissertation.

Chapter 3 introduces benchmark environments designed to systematically introduce novelty and evaluate agent behavior under open-world conditions. These environments provide mecha-

nisms for controlled novelty injection and enable rigorous evaluation of how agents detect and respond to unexpected changes in their environment.

Chapter 4 develops adaptive planning–learning systems that allow agents to recover from execution failures during task execution. The chapter presents approaches that integrate planning mechanisms with learning-based adaptation, enabling agents to detect when previously learned models become insufficient and modify their behavior in response to unexpected situations.

Chapter 5 investigates how robots can acquire interaction policies that generalize across objects and tasks. It presents object-centric and force-based reinforcement learning approaches for manipulation tasks involving sustained contact and complex object dynamics, demonstrating how these methods enable robots to learn interaction strategies that transfer across diverse environments and manipulation scenarios.

Chapter 6 concludes the dissertation by summarizing the main findings and discussing future directions for research on autonomous agents and robots operating in open-world environments.

Chapter 2

Background

This chapter provides the background necessary for the remainder of the dissertation. It is organized in two parts. The first part surveys prior work on open-world autonomy, situating this thesis within the broader landscape of AI and robotics research. The second part introduces the formal frameworks that underpin the technical contributions of the dissertation.

The dissertation is grounded in the classical agent–environment paradigm, in which an agent perceives its surroundings, reasons and plans over possible actions, and executes those actions to achieve its goals. This perceive–plan–execute loop, and its natural extension to include adaptation and recovery, provides a unifying lens through which to examine the challenges of open-world autonomy. In closed-world settings, each stage of this loop has been studied extensively and with considerable success. However, when agents are deployed in open-world environments, the assumptions underlying each stage are violated: new objects may appear, dynamics may change, and actions may fail in unforeseen ways. Critically, novelty at any stage of the loop can propagate failures across the others, making open-world autonomy a systemic challenge rather than an isolated one.

Section 2.1 reviews prior work along each stage of the agent loop under open-world conditions – sensing, planning and learning, and execution and recovery—and identifies the gaps that motivate the contributions of this dissertation. Section 2.2 then introduces the formal models used throughout: symbolic planning (Section 2.2.1), reinforcement learning (Section 2.2.2), hybrid planning and learning systems (Section 2.2.3), and a formalization of novelty relative to an agent's knowledge and models (Section 2.2.4).

2.1 Related Work

This section reviews prior work relevant to each stage of the perceive–plan–execute loop under open-world conditions. Rather than providing exhaustive coverage of any single area, the discussion is organized around three pillars of open-world autonomy—sensing and perception, planning and learning, and execution and recovery—and aims to identify the key challenges and limitations that motivate the contributions of this dissertation.

2.1.1 Open-World Sensing and Perception

In closed-world settings, perception systems are designed to recognize a fixed set of categories, objects, or states defined at training time. Open-world perception, by contrast, requires the ability to detect and reason about entities and situations that were not anticipated during system design. Early work on open set recognition [28, 3] extended standard classifiers to reject inputs that do not belong to any known category, typically through confidence thresholding or distribution boundary modeling. Out-of-distribution detection methods [16] further developed principled approaches for identifying inputs that fall outside the training distribution.

More recently, vision-language models (VLMs) have transformed the landscape of open-world perception. CLIP [26] demonstrated that joint vision-language pretraining enables zero-shot recognition of arbitrary categories described in natural language, fundamentally relaxing the closed-vocabulary constraint. This capability has been extended to object detection through open-vocabulary detectors such as ViLD [11] and OWL-ViT [23], which can localize objects described by free-form text queries without category-specific training. Foundation models for segmentation, such as the Segment Anything Model [19], further demonstrate that general-purpose visual representations can support open-ended perception tasks.

Despite these advances, a critical gap remains between perceptual novelty detection and the needs of an acting agent. Recognizing that an object is unfamiliar or that a scene has changed is necessary but not sufficient: the agent must also determine whether the novelty is functionally relevant to its current task and how it should modify its behavior in response. Current perception systems largely operate as passive classifiers and do not reason about the implications of detected novelty for downstream planning and execution.

2.1.2 Open-World Planning and Learning

Planning and learning provide the reasoning and decision-making capabilities that allow agents to select actions in pursuit of goals. Classical symbolic planning, formalized through STRIPS [6] and PDDL [22], enables structured long-horizon reasoning by searching over sequences of operators with well-defined preconditions and effects. Modern planners such as Fast Downward [15] achieve strong performance across diverse domains through domain-independent heuristics. However, symbolic planning assumes that the domain model is complete and correct; when the model is inaccurate or missing operators, generated plans may fail during execution.

Reinforcement learning offers a complementary approach, enabling agents to learn effective behaviors through interaction without requiring explicit models. Deep RL algorithms, including DQN [24], PPO [30], and SAC [13], have demonstrated success in complex control tasks. Model-based approaches, including world models such as Dreamer [14] and planning-based methods such as MuZero [29], learn environment dynamics and use them for decision-making. These methods reduce the need for hand-specified models but typically assume that the learned dynamics remain stationary.

A recent and rapidly growing line of work leverages large pretrained models for robotic planning and decision-making. Vision-language-action (VLA) models such as RT-2 [34] directly map visual observations and language instructions to robot actions, drawing on the broad world knowledge encoded in large language and vision models. SayCan [1] and related approaches [17] use large language models as high-level planners that decompose natural language instructions into sequences of learned skills. Generalist robot policies such as Octo [32] and π_0 [4] aim to learn cross-embodiment policies from large-scale multi-robot datasets. While these approaches demonstrate impressive breadth, they face challenges related to physical grounding, hallucination of infeasible actions, and robustness to environmental conditions that diverge from their training data. In open-world settings, where the nature of the change may be qualitatively different from anything in the training distribution, the reliability of these systems remains an open question.

2.1.3 Open-World Execution and Recovery

Even when perception and planning succeed, action execution in open-world environments may fail due to unexpected dynamics, novel object properties, or environmental changes. Recovery from such failures requires mechanisms that go beyond standard error handling: the agent must identify why the failure occurred and adapt its behavior accordingly.

Classical approaches to execution-time adaptation include plan monitoring and repair [7, 8], which detect deviations from expected execution outcomes and revise plans within the existing domain model. However, plan repair assumes that the domain model remains valid and that a correct plan exists within the current model, which may not hold when qualitative novelty has occurred. Continual learning methods [20, 27] address the problem of acquiring new capabilities without forgetting previous ones, but most assume known task boundaries and distributions. Sim-to-real transfer techniques, including domain randomization [33] and system identification [25], improve policy robustness by training across varied conditions, but the variations are typically parameterized in advance and may not cover the kinds of qualitative changes encountered in open-world settings.

More recently, work within the DARPA SAIL-ON program [31] has specifically targeted the problem of agents that must detect, characterize, and accommodate novelty in their environments. This program motivated research on novelty as an agent-relative concept: what constitutes novelty depends on the agent's prior knowledge and capabilities, not solely on properties of the environment [5, 21]. Several approaches have explored how agents can adapt their policies or models in response to detected novelty [2, 10], though systematic methods for efficient recovery from qualitatively new failure modes remain limited. Some efforts have sought to address multiple stages of the open-world agent loop in an integrated manner. World models [12, 14] offer a potential unifying framework by learning environment dynamics that can serve both perception and planning, though they typically do not handle discrete, qualitative changes in the environment. End-to-end foundation models for robotics [34, 32] implicitly combine perception, reasoning, and action selection in a single model, but their behavior under genuine novelty is not well understood. Task and motion planning systems [9, 18] integrate symbolic and geometric reasoning but inherit the brittle model assumptions of classical planning. De-

spite growing interest in open-world autonomy, the field remains largely fragmented: methods that address perception under novelty, planning under incomplete models, and recovery from execution failures are typically developed in isolation from one another.

The preceding discussion has examined open-world autonomy through the lens of the perceive–plan–execute loop, identifying challenges at each stage: sensing under novelty, planning and learning with incomplete models, and execution and recovery when prior knowledge becomes insufficient. Drawing primarily from the challenges identified in planning and learning (Section 2.1.2) and execution and recovery (Section 2.1.3), this dissertation makes progress on three interconnected challenges. First, the lack of standardized environments for systematically evaluating agents under controlled novelty limits rigorous comparison of approaches and slows progress in open-world AI. Second, existing hybrid planning-learning systems do not adequately address novelty handling: they lack mechanisms for detecting execution failures caused by qualitative environmental changes, learning recovery behaviors, and reusing acquired knowledge across subsequent encounters with novelty. Third, robotic skill learning methods remain limited in their ability to produce interaction policies that generalize across objects and platforms, particularly for contact-rich manipulation tasks.

These three challenges correspond to the core contributions of this dissertation. Chapter 3 introduces benchmark environments for evaluating open-world agents under controlled novelty injection. Chapter 4 develops hybrid planning-learning architectures that enable agents to adapt and recover when previously learned models become insufficient. Chapter 5 presents object-centric, force-based reinforcement learning methods for robotic manipulation that generalize across objects and robot embodiments. Together, these contributions advance the goal of building autonomous agents and robots that can operate robustly in open-world environments where novelty is not the exception but the norm.

2.2 Preliminaries

2.2.1 Symbolic Planning

Symbolic planning models perform sequential decision-making using an explicit representation of states, actions, and goals through symbols. The environment is described using a first-

order language consisting of predicate symbols and function symbols. Predicate symbols define Boolean properties of objects, while function symbols define numeric quantities.

A fluent is a state variable whose value may change over time. Boolean fluents are obtained by grounding predicate symbols with objects in the domain, and numeric fluents are obtained from function symbols. Let $\mathcal{F}$ denote the set of all fluents. A symbolic state $\tilde{s} \in \tilde{\mathcal{S}}$ is defined as an assignment of values to the fluents in $\mathcal{F}$, assigning each Boolean fluent a value in $\{0, 1\}$ and each numeric fluent a value in $\mathbb{R}$. In propositional domains, a symbolic state can be equivalently represented as the set of grounded predicates that are true, with all others assumed false under the closed-world assumption.

The symbolic model is defined by a finite set of operators $\mathcal{O}$. Each operator $o \in \mathcal{O}$ is defined as $o = \langle \psi_o, \omega_o \rangle$, where ψ_o is a logical formula over fluents specifying when the operator is applicable, and ω_o specifies how the state changes after execution. An operator o is applicable in a state $\tilde{s}$ if $\tilde{s} \models \psi_o$. Effects modify Boolean fluents through add and delete operations and update numeric fluents through assignments or arithmetic expressions. Operators are typically defined as parameterized schemas and instantiated over objects to produce grounded operators.

A planning task is defined as $\mathcal{T} = \langle \mathcal{F}, \mathcal{O}, \tilde{s}_0, g \rangle$, where $\tilde{s}_0 \in \tilde{\mathcal{S}}$ is the initial state and g is a logical condition over fluents. A plan is a finite sequence of operators $\rho = [o_1, \ldots, o_n]$. Executing a plan from the initial state produces a sequence of symbolic states obtained by sequentially applying applicable operators. A plan is valid if execution from $\tilde{s}_0$ results in a state that satisfies g. Classical symbolic planning assumes deterministic transitions, where the effects of an operator uniquely determine the resulting state.

Planning domains and problems are commonly specified using the Planning Domain Definition Language (PDDL), which provides a standard representation for predicates, functions, operators, and planning tasks.

2.2.2 Reinforcement Learning

Reinforcement learning models sequential decision-making in which an agent learns to select actions through interaction with an environment so as to maximize cumulative reward. At each time step, the agent observes a state, selects an action, receives a reward signal, and transitions to a new state.

Reinforcement learning problems are modeled as a Markov Decision Process (MDP), defined as $M = \langle \mathcal{S}, \mathcal{A}, P, R, \gamma \rangle$, where $\mathcal{S}$ is the set of states, $\mathcal{A}$ is the set of actions, $P(s' \mid s, a)$ defines the probability of transitioning to state s' after taking action a in state s, $R(s, a)$ is the reward function, and $\gamma \in [0, 1)$ is the discount factor.

At time step t, the agent is in state $s_t \in \mathcal{S}$, selects an action $a_t \in \mathcal{A}$, receives reward $r_{t+1} = R(s_t, a_t)$, and transitions to a new state s_{t+1} according to $P(\cdot \mid s_t, a_t)$. The Markov property assumes that the transition and reward depend only on the current state and action.

A policy $\pi(a \mid s)$ defines the behavior of the agent by specifying a probability distribution over actions given a state. The objective of reinforcement learning is to find a policy that maximizes the expected cumulative reward.

The return from time step t is defined as $G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$. The state-value function is defined as $V^\pi(s) = \mathbb{E}_\pi[G_t \mid s_t = s]$, and the action-value function is defined as $Q^\pi(s, a) = \mathbb{E}_\pi[G_t \mid s_t = s, a_t = a]$.

These value functions satisfy recursive relationships known as the Bellman expectation equations. The state-value function satisfies

$$V^\pi(s) = \mathbb{E}_\pi \left[r_{t+1} + \gamma V^\pi(s_{t+1}) \mid s_t = s \right],$$

and the action-value function satisfies

$$Q^\pi(s, a) = \mathbb{E}_\pi \left[r_{t+1} + \gamma \mathbb{E}_{a' \sim \pi(\cdot \mid s_{t+1})} Q^\pi(s_{t+1}, a') \mid s_t = s, a_t = a \right].$$

In discrete settings, the state and action spaces are finite, i.e., $|\mathcal{S}| < \infty$ and $|\mathcal{A}| < \infty$. In this case, value functions such as $V^\pi(s)$ and $Q^\pi(s, a)$ can be represented explicitly in tabular form, and reinforcement learning algorithms can directly estimate these quantities for each state or state-action pair.

In continuous settings, the state space $\mathcal{S} \subseteq \mathbb{R}^n$ and/or the action space $\mathcal{A} \subseteq \mathbb{R}^m$ may be continuous. In such cases, value functions and policies cannot be represented exactly and are instead approximated using parameterized function approximators, typically implemented using neural networks. A policy is represented as $\pi_\theta(a \mid s)$, where θ denotes the parameters

of the policy, and value functions are represented as $V_\phi(s)$ and $Q_\phi(s, a)$, where ϕ denotes the parameters of the function approximator.

Learning in continuous settings involves optimizing the parameters θ and ϕ to maximize expected return. These formulations are particularly important in robotics and control tasks, where states and actions are naturally continuous and policies must operate over high-dimensional spaces.

2.2.3 Hybrid Planning and Learning

Symbolic planning and reinforcement learning provide complementary approaches to sequential decision-making. Symbolic planning enables structured reasoning over high-level representations but assumes accurate and complete models of the environment. Reinforcement learning enables agents to learn behaviors through interaction but typically operates over lower-level or subsymbolic state representations and may lack explicit structure. In many settings, reinforcement learning can also be viewed hierarchically, where high-level symbolic operators correspond to abstract actions, and their execution is realized through lower-level policies or planners. Hybrid planning and learning systems aim to combine these paradigms by integrating symbolic reasoning with learned or executed behaviors.

In hybrid systems, decision-making is defined over two representations. The symbolic model is represented by a planning task $\mathcal{T} = \langle \mathcal{F}, \mathcal{O}, \tilde{s}_0, g \rangle$,and the execution is formalized by a Markov decision process $M = \langle \mathcal{S}, \mathcal{A}, P, R, \gamma \rangle$. The symbolic model operates over discrete representations, whereas the execution model operates over environment states that may be continuous or stochastic. An abstraction function $\alpha : \mathcal{S} \to \tilde{\mathcal{S}}$ maps environment states to symbolic states.

Operators in the symbolic model correspond to executable behaviors in the environment. Each operator $o \in \mathcal{O}$ is associated with a policy $\pi_o(a \mid s)$ that defines its execution in the environment. A symbolic plan $\rho = [o_1, \ldots, o_n]$ therefore induces a sequence of policies executed in the environment.

A hybrid planning and learning system is defined as $\mathcal{H} = \langle \mathcal{T}, M, \alpha \rangle$, which formalizes the separation between symbolic decision-making over abstract representations and execution in the environment.

2.2.4 Novelty in Open-World Environments

Novelty is defined relative to an agent's knowledge. An element of the environment is considered novel if it cannot be represented or inferred using the agent's current models of the world.

Let the agent's knowledge be defined by a symbolic model $\mathcal{T} = \langle \mathcal{F}, \mathcal{O}, \tilde{s}_0, g \rangle$ and an execution model $M = \langle \mathcal{S}, \mathcal{A}, P, R, \gamma \rangle$. The symbolic model specifies structured knowledge over abstract states and operators, while the execution model specifies the dynamics of interaction with the environment.

A novelty corresponds to the introduction of elements that are not contained in the agent's current knowledge. These may include new fluents, operators, states, actions, or changes in transition dynamics. Formally, a novelty induces an extension of the agent's model, such as $\mathcal{F}'$, $\mathcal{O}'$, $\mathcal{S}'$, or $\mathcal{A}'$, where at least one of the following holds:

$$\mathcal{F}' \cap \mathcal{F} = \emptyset, \quad \mathcal{O}' \cap \mathcal{O} = \emptyset, \quad \mathcal{S}' \not\subseteq \mathcal{S}, \quad \mathcal{A}' \not\subseteq \mathcal{A}.$$

In hybrid planning and learning systems, novelty may manifest as a discrepancy between symbolic reasoning and execution. In particular, an operator $o \in \mathcal{O}$ may be applicable in a symbolic state $\tilde{s}$, i.e., $\tilde{s} \models \psi_o$, but its execution in a corresponding environment state may fail to produce the intended effects. Such discrepancies arise when the agent's models are incomplete with respect to the true environment.

This formulation captures novelty as a deviation between the agent's current knowledge and the environment, and provides a basis for reasoning about adaptation in open-world settings.

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